

## Assignment 01: Part-of-Speech (POS) Extraction from a News Article

### Objective:

Apply Natural Language Processing techniques to identify and extract nouns and verbs from an English news article.

### Tasks

1. Download any **English-language news article** from a reputable news source.
2. Save the article content as a **plain text (.txt) file**.
3. Load the text file into a Python program or Jupyter Notebook.
4. Perform **Part-of-Speech (POS) tagging** on the text.
5. Extract all **Nouns** and **Verbs** from the article.
6. Save the extracted words into a **CSV file** with appropriate columns (e.g., Word, POS\_Tag).
7. Submit your solution as a Jupyter Notebook.

### Submission Requirements

- **Notebook File Name Format:**  
`YourName_POS_Identifier.ipynb`

#### Example:

`PremBist_POS_01.ipynb`

- **Output File:**  
`YourName_POS_Identifier.csv`

### Deliverables

- Jupyter Notebook (.ipynb)
- Original news article text file (.txt)
- CSV file containing the extracted nouns and verbs (.csv)

### Evaluation Criteria

- Correct loading and processing of the text file
- Accurate POS tagging and extraction of nouns and verbs
- Proper CSV export
- Code readability, documentation, and organization
- Adherence to the required file naming convention

## Assignment 02: Named Entity Recognition (NER) from a News Article

## Objective:

Apply Named Entity Recognition (NER) techniques to identify and extract named entities from an English news article.

## Tasks

1. Download any **English-language news article** from a reputable news source.
2. Save the article content as a **plain text (.txt) file**.
3. Load the text file into a Python program or Jupyter Notebook.
4. Perform **Named Entity Recognition (NER)** on the text using an NLP library such as **spaCy, NLTK, or Hugging Face Transformers**.
5. Extract all named entities and their corresponding entity types (e.g., PERSON, ORG, GPE, DATE, MONEY, etc.).
6. Save the extracted entities into a **CSV file** with appropriate columns (e.g., Entity, Entity\_Type).
7. Provide a summary of the different entity types found in the article.
8. Submit your solution as a Jupyter Notebook.

## Submission Requirements

- **Notebook File Name Format:**  
YourName\_NER\_Identifier.ipynb

**Example:**  
PremBist\_NER\_01.ipynb

- **Output File:**  
YourName\_NER\_Identifier.csv

## Deliverables

- Jupyter Notebook (.ipynb)
- Original news article text file (.txt)
- CSV file containing the extracted named entities (.csv)
- A brief summary of entity counts by category

## Example CSV Format

| Entity        | Entity_Type |
|---------------|-------------|
| Donald Trump  | PERSON      |
| United States | GPE         |
| Microsoft     | ORG         |
| July 20, 2026 | DATE        |

## Evaluation Criteria

- Correct loading and preprocessing of the text file
- Accurate implementation of Named Entity Recognition
- Proper extraction and labeling of entities
- Correct CSV export and formatting
- Quality of code, documentation, and analysis
- Adherence to the required file naming convention

**Bonus (Optional):**

- Visualize the recognized entities using spaCy's `displacy` or another visualization tool.
- Compare the performance of two different NER models and discuss the results.